midterm

Graded

Student

Atharva Chalke

Total Points

86 / 100 pts

Question 1

p1

Resolved **10** / 15 pts

- ✔ **− 5 pts** may accept odd 0s (or reject even 0s)
- ✔ **− 5 pts** may accept even 1s (or reject odd 1s)
- **− 5 pts** Not a DFA
- **− 15 pts** Not attempted or non-relevant answer
- **− 0 pts** Complete
- **− 1 pt** minor issue
- **− 2.5 pts** accepts empty string
- **− 4 pts** Close, but you got two edge labels wrong! e.g. try 110
- **− 1 pt** extra final state!
- ✔ **+ 5 pts** showed understanding and effort but din't reach correct answer.

Regrade Request **Submitted on: May 22**

Regrade Requested (NFA AND DFA MADE FOR Even 0s or Odd 1s instead of Even 0s and odd 1s).

+5

Reviewed on: May 24

Question 2

p2

Resolved 37 / 40 pts

- **0 pts** Correct
- **30 pts** missing essential logical steps in the proof - see solutions.
- **5 pts** Need more detail on why S' is accepted by the DFA
- **5 pts** Need more detail on why S' is not in the language
- **2 pts** Need to say that the loop is in the computation history.
- **20 pts** The selection of S' does not work. You can't arbitrarily set x and z.
- ✔ **3 pts** need to be clearer on why S' is accepted by the DFA
- **10 pts** Can't just define S as xyz without defining y first, using computation loops.
- **2 pts** We can't conclude that xy^iz is in L. We only can conclude that M would accept it.
- **5 pts** vague details at several points
- **40 pts** Not answered
- **38 pts** a regular language can contain a subset of strings that do not form a regular language. For example Sigma* contains the palindromes. So your basic approach will not work. The problem asked for the template proof.
- **15 pts** need to argue that S' is accepted by M.
- **5 pts** need to be clearer on why S' is not in L.
- **40 pts** see solutions

1 This needs more justification. See the solutions.

Regrade Request **Submitted on: May 17**

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please see the added comment in the body of the exam.

Reviewed on: May 29

Question 3

p3 **19** / 25 pts

- **0 pts** Correct
- **1 pt** start state is 1
- **1 pt** {1} , a --> {2}
- **2 pts** {1} , b --> {1,2}
- **2 pts** {2} , a --> {1,3}
- **1 pt** {2} , b --> {2}
- ✔ **2 pts** {1, 2} , a --> {1, 2, 3}
- ✔ **2 pts** {1, 2} , b --> {1, 2}
- **2 pts** {1, 3} , a --> {2}
- ✔ **2 pts** {1, 3} , b --> {1, 2}
- **2 pts** {1, 2, 3} , a --> {1, 2, 3}
- **2 pts** {1, 2, 3} , b --> {1, 2}
- **2 pts** {2} is accept state
- **2 pts** {1, 2} is accept state
- **2 pts** {1, 2, 3} is accept state
- **2 pts** {1, 3} is not an accept state
- **2 pts** {1} is not an accept state

Question 4

p4 **20** / 20 pts

- ✔ **0 pts** Correct
- **6 pts** 1st part: The grammar is context-free.
- **6 pts** 1st part: no answer
- **3 pts** 1st part: There should be an explanation or the explanation is incorrect
- **6 pts** 2nd part: The language is palindrome
- **3 pts** 2nd part: There is no explanation why it is palindrome or the explanation is incorrect
- **5 pts** 3rd part: It generates more than two c.
- **3 pts** 3rd part: It doesn't generate all possible strings (e.g. "abcbaabcba" will not be generated)
- **8 pts** 3rd part: No (proper) answer
- **8 pts** 3rd part: It doesn't generate a palindrome language with {a, b} and one pair of {c}
- **5 pts** 3rd part: It generates strings without any c.
- **2 pts** 3rd part: This grammar does not obey the context-free definition.

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CS 120 Spring 2018. Midterm, May 6, 2018

INSTRUCTIONS:

a. Write your name of EACH page, and if you use the back of a page, write your name there also. We will be separating the pages and scanning them, so if your name is not on a given page, or on a side where you wrote an answer, you might not get credit for your answer.

b. Before the exam starts, ask the person to your right and to your left what their first and last names are, and write those names here:

Carzarnere Connie ---------- Your left neighbor

---------- Your right neighbor

c. For each question, write your answer on the page where the question is stated. You may use the back if needed. You may ask for scratch paper if you need it.

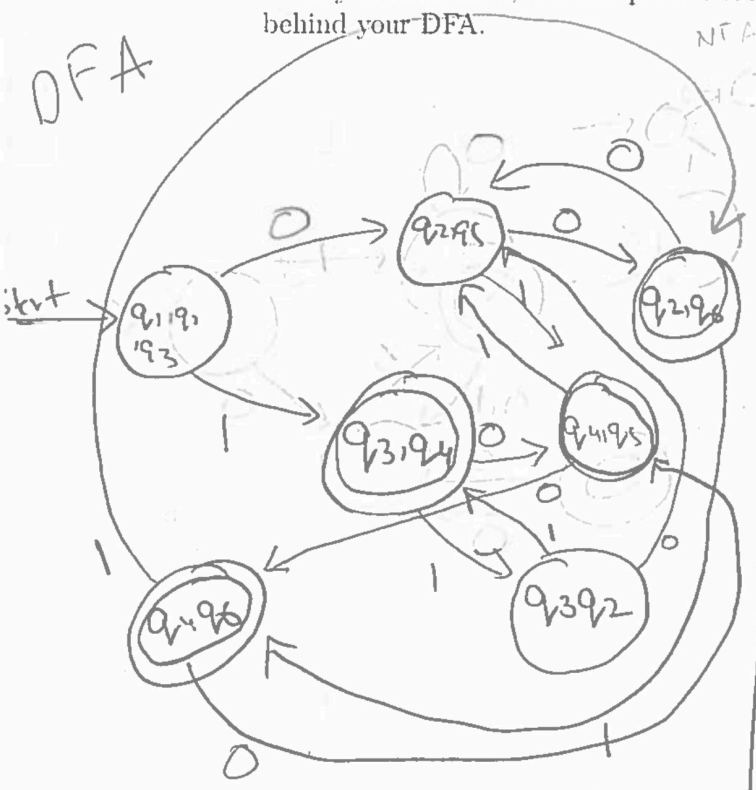
d. Try to be brief and to the point.

Start of Problems: The time listed for each problem is an estimate of how much time a super-prepared student would need for that problem.

1. (5 minutes) Create a DFA to show that the language L = set of strings with an even number of 0s and an odd number of 1s, where the alphabet $\Sigma = \{0, 1\}$. Show the state diagram for your DFA.

If you have time, and for partial credit in case your details are wrong, explain the idea behind your DFA.

DFA



NFA

State	0	1
q1,q2,q3	q2,q5	q4,q3
q2,q5	q2,q6	q4,q5
q3,q4	q4,q5	q3,q2
q2,q6	q2,q5	q4,q6
q4,q5	q4,q6	q2,q5
q3,q2	q2,q5	q3,q4

	0	1
q4 q6	q4,q5	q2,q6

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2. (15 mins) For $\Sigma = \{a, b\}$, and a string $w \in \Sigma^*$, let $\overline{w}$ denote the string obtained fro w be replacing every occurrance of 'a' with 'b', and every occurrance of 'b' with 'a'. Fc example, if $w = abbab$ then $\overline{w} = baaba$. Now let $L = \{w\overline{w} : w \in \Sigma^*\}$. That is, L consist of a string $w \in \Sigma^*$ followed by $\overline{w}$. Write out the full template proof to show that L is not regular language.

Hint: A suggested string you can use for S has the form: a run of character 'a' followe by a run of character 'b'. A hint for later in the proof: If a string v only has $a's$ in it, an an equal-length string u has any $a's$ in it, then the string vu cannot be in L.

$\Sigma = \{a, b\}$

Assume $L = \{\omega\overline{\omega}$ is a regular language$\}$

Lets look at Machine M with p states.

Let ~~$S = a^p$~~ $S = \overline{a^{p+1} b}\ \omega\overline{\omega}$ s.t. $\omega = a^{p+1}$ $\therefore \overline{\omega} = b^{p+1}$

$S = a^{p+1} b^{p+1}$

Since length of strings of $\underline{\omega}$ is greater then number of states in M, there will be at least one loop in $\underline{\omega}$

$\therefore$ Let the ~~point~~ state which occurs ~~twice~~ at least twice in a loop be a_k.

$\therefore$ A ~~pt~~ for string S, we have a section before a_k, a section that loops atleast once to a_k, and a section that computes the rest of ω and then ω^r. Let X be ~~Let~~ ~~X~~ the first part, Y the second part and Z the third part.

$\therefore$ $X = a \ldots a$

$Y = a \ldots a$

$Z = a \ldots abbb \ldots bbb$

PTO

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Since Y loops, for $S = XYZ$, $S' = XYYZ$ will also be accepted by the machine.

for $\hat{\omega}\overline{\omega}$ we will have same length on both sides.

$\therefore$ on pumping up, mid point shifts left.

Now since the left side contains all a's, the right side should contain all b's; but, in this case, there will be atleast one instance of ba on the right.

Therefore, this language is not in the language, but is accepted by M.

Hence, our initial assumption is False.

$\therefore$ L is not a regular language

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3. (10 mins) Use the algorithm learned in class to convert the NFA shown in Figure 1 to a DFA. You don't need to show the intermediate steps unless you want to. But you need to keep your answer on this one sheet of paper (front and back).

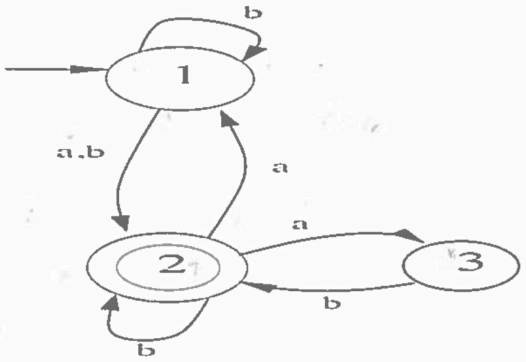
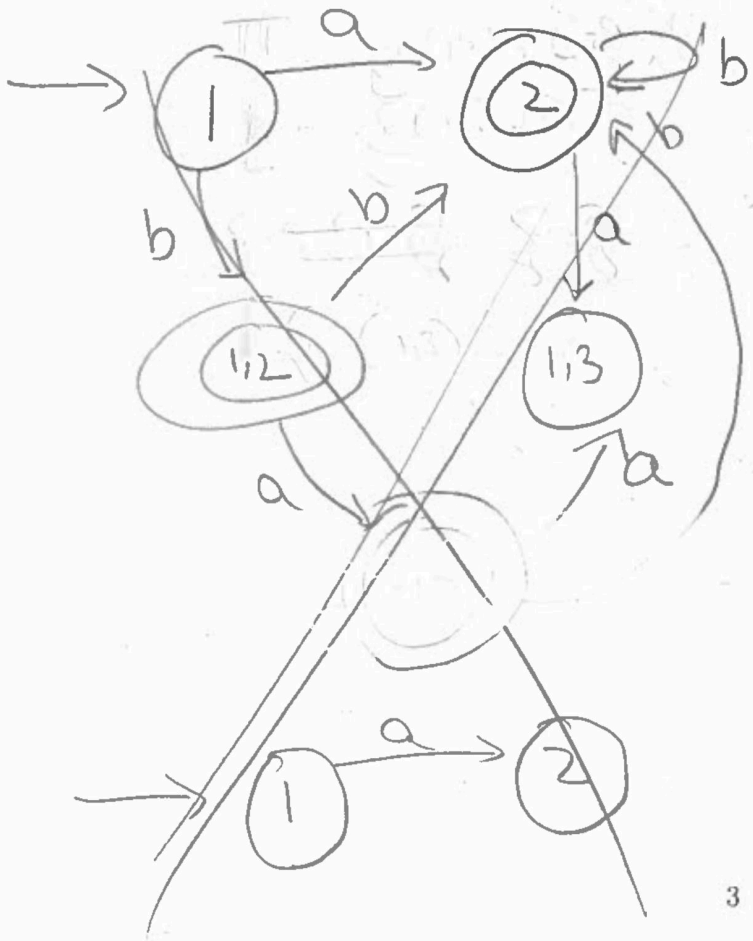


Figure 1: The NFA for problem 4a



	a	b
1	2	(1,2)
2	1,3	2
1,~~2~~	2,~~3~~	2
1,3	2	1,2,3
~~2,3~~	~~1,3~~	~~2~~
1,2,3	2,3,1	1,2

P.T.O

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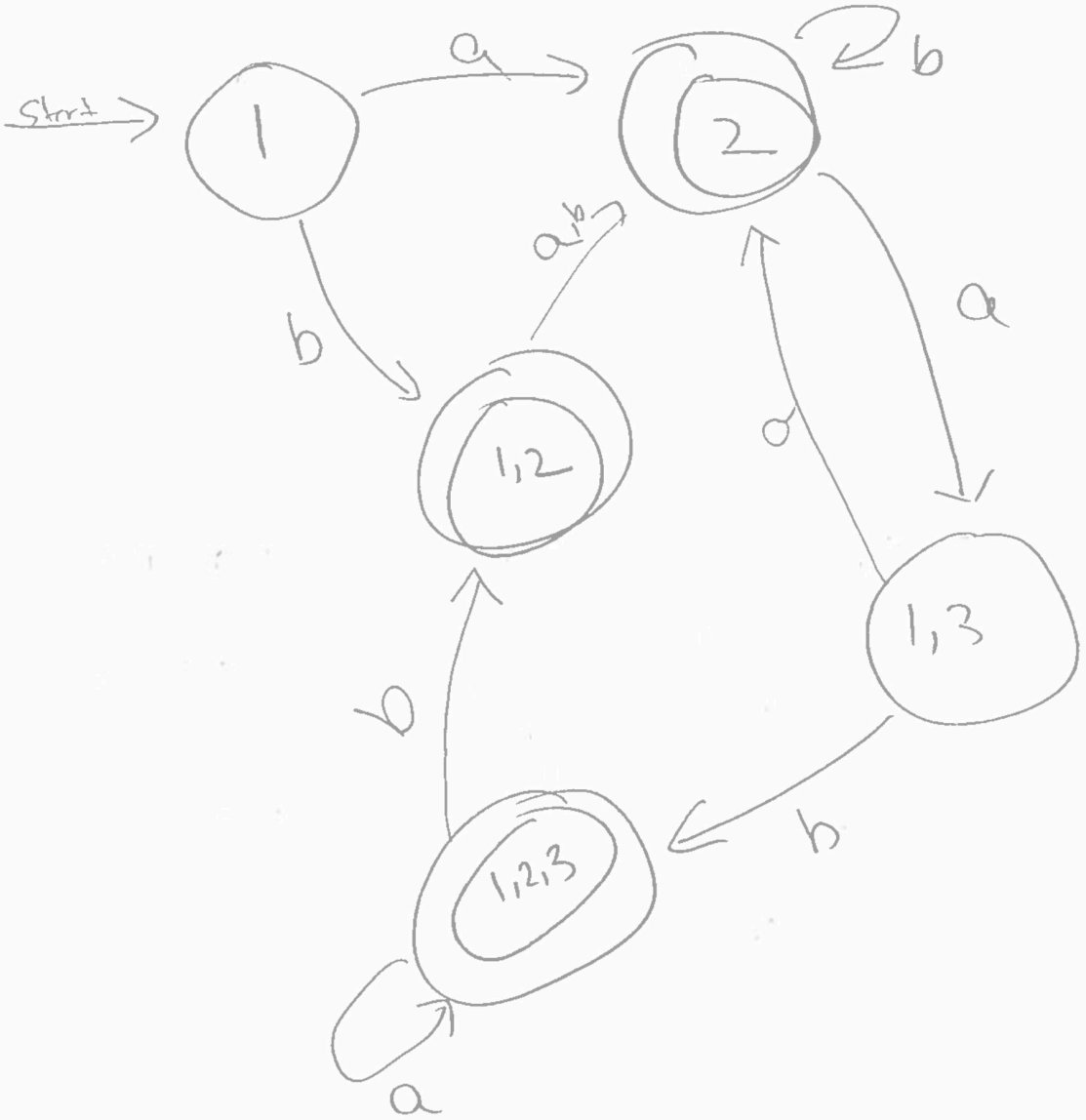

Atharva Chelke.

4. (10 mins) Is the following grammar Context-Free? Explain. The only non-termina symbol is S. The terminal symbols are $\{a, b, c\}$, and the productions are:

$S \rightarrow aSa$
$S \rightarrow bSb$
$S \rightarrow cSc$
$S \rightarrow \epsilon$

Does this grammar generate the language of palindromes with the alphabet $\Sigma = \{a, b, c\}$ Explain.

Modify the grammar so that the strings it generates never have more than two occurrance of the character 'c'.

Yes since there is only one alphabet on leftside.

Yes, this geammar ~~B~~ generates a palindrome, since for every character on the leftside, ~~th~~ of S, there is the same ~~a~~ charac on the right side of S. ∴ it will always be balanced and in reverse order.

Modification

$S \rightarrow aDa \mid bDb \mid cDc$
$D \rightarrow aDa \mid bDb \mid cEc$
$E \rightarrow aEa \mid bEb \mid \varepsilon$
$S \rightarrow \varepsilon$

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